UN-NAFEU ROHIT

BSc. IN EEE

Kazipara, Dhaka
unnafeurohit@gmail.com
unnafeurohit.com
+880 16002-00008

CAREER OBJECTIVE

Electrical and Electronic Engineering graduate with expertise in embedded systems, IoT, PCB design, wireless communication, and machine learning. Experienced in developing intelligent hardware solutions through research, industrial training, and IEEE-published projects. Seeking an entry-level Hardware, Embedded Systems, IoT, or R&D Engineering role where I can apply my technical expertise, analytical thinking, and problem-solving skills to develop innovative and reliable engineering solutions.

ACADEMIC QUALIFICATION

- **MSc in Applied Physics & Electronics**
  Jahangirnagar University
  (Expected Completion: 2027)
- **BSc in Electrical and Electronic Engineering**
  Daffodil International University (2022–2025)
- **Secondary School Certificate (SSC)**
  Police Lines School & College,Rangpur (2018)
- **Higher Secondary School Certificate (HSC)**
  Police Lines School & College,Rangpur (2020)

TECHNICAL SKILLS

- **Simulation & Design Tools:** Proteus, MATLAB, Simulink, Fritzing, CST, AutoCAD 2D.
- **PCB Design:** Schematic Capture, PCB Layout, Gerber Generation (EasyEDA, Proteus).
- **Programming Languages:** C, C++,Python.
- **Hardware & IoT Platforms:** Arduino IDE, ESP32, Raspberry Pi (basic), Pixhawk (basic).
- **IoT Project Design:** End-to-end IoT system development, sensor integration, and wireless communication (LoRa, Wi-Fi, etc.).
- **Circuit Design & Prototyping:** Analog/digital circuit design, breadboard prototyping, and hardware implementation.
- **Lab & Instrumentation:** Skilled in oscilloscopes, function generators, and electrical measurement tools

PUBLICATIONS

J. Shariar, R. Rayhan, U. N. Rohit, H. M. R. Ahammed, and T. Fouzder, **"A solar-powered IoT system for real-time water quality monitoring using LoRa and machine learning,"** in Proc. 2025 **IEEE 2nd Int. Conf. Computing, Applications and Systems (COMPAS),** Kushtia, Bangladesh, Oct. 2025, pp. 1-6, *doi: 10.1109/COMPAS67506.2025.11381845*.

EXPERIENCE

Industrial Trainee — Tools and Technology Institute (TTI), BITAC | January 12 – February 10, 2025
Completed hands-on industrial training on Mechatronics & PLC

ACADEMIC & TECHNICAL PROJECTS

- **ML-Based Intelligent Environmental Quality Prediction System - Final Year Project**
  Developed a solar-powered IoT system using LoRa communication and machine learning to predict and classify environmental quality in real time for remote and disaster-prone areas.
- **Solar-Powered IoT Water Quality Monitoring System Using LoRa and Machine Learning (2025)**
  Designed an energy-autonomous IoT platform integrating water quality sensors, long-range communication, and ML-based real-time classification.
- **Real-Time Intelligent Fault Classifier of Three-Phase Induction Motor (ML-Based) (2025)**
  Implemented a machine learning model for real-time fault detection and classification of three-phase induction motors using electrical parameters.
- **Performance Analysis of Photovoltaic Panel Using Machine Learning (2025)**
  Analyzed photovoltaic performance and efficiency trends under varying environmental conditions using supervised machine learning techniques.
- **Metamaterial-Based Multiband Antenna Design (5 / 7.5 / 10.5 GHz) Using CST (2025)**
  Designed and simulated a metamaterial-based multiband antenna in CST Microwave Studio, achieving resonances at 5 GHz, 7.5 GHz, and 10.5 GHz with improved bandwidth and radiation characteristics.
- **Secure PIN-Based Smart Relay Control System Using ESP32 and Web Dashboard (2025)**
  Built a secure load control system with PIN authentication, ESP32-based control, and real-time monitoring through a web dashboard.

- **Smart Energy Meter Using ESP32 and Blynk Dashboard (2024)**
  Developed a smart energy monitoring system with real-time visualization and remote access using a cloud-based dashboard.
- **PLC-Based Temperature Control System Design (2024)**
  Designed and implemented an industrial temperature control system using PLC logic and sensor feedback.
- **Overheat Protection System for Electric Motor (2024)**
  Implemented a safety system that automatically disconnects motors during overheating conditions to prevent damage.
- **Full Flat Layout Electrical Design Using AutoCAD (2022)**
  Prepared a complete residential electrical layout including lighting, power outlets, and protection devices.

A C H I V E M E N T S

- 1st Position, National Science Olympiad (Division Level), 2014
- 2nd Position in Salim Al Deen National Art Competition, 2017

I N T E R E S T S

- Consumer Electronics and Computer Technology
- Artificial Intelligence and Machine Learning
- IoT and Robotics
- Esports

R E F E R E N C E

- **Dr. Tama Fouzder**
  Associate Professor
  Department of Electrical and Electronic Engineering
  Daffodil International University
  Dhaka, Bangladesh
  Email: tamafouzder.eee@diu.edu.bd
  Phone: (+880)1747744721
- **Dr. Md. Rezwanul Ahsan**
  Associate Professor
  Department of Electrical and Electronic Engineering
  Daffodil International University
  Dhaka, Bangladesh
  Email: ahsan.eee@diu.edu.bd
  Phone: (+880)1712-599409